Chapter 5: Financial Statement Analysis - Ratios

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Plus: Altman Z-Score • Working Capital Analysis • Key Ratio Formulas

Extra: Using Ratios Analyzing and Interpreting Core Operating Activities

Part 1

Vertical & Horizontal Analysis

Vertical & Horizontal Analysis

What is Financial Statement Analysis?

Identifies **relationships** and **trends** within financial statements **over time** and **compared to other companies** or industry averages.

Who Uses It?

Investors
Assess investment risk and growth potential

Creditors
Evaluate credit risk and repayment ability

Managers
Monitor performance and make decisions

Before Analyzing — Understand the Business Context

Life cycle stage: Start-up, growth, or mature?

Competitive position: How does it compare to rivals?

Capital structure: Reliance on debt vs. equity?

Regulatory environment: Laws affecting operations?

Vertical Analysis – Common-Size Statement

Expresses each line item as **a percentage** of a designated **base amount**

Income Statement

Base = Net Sales (100%)

Net Sales	100%
COGS	45%
Gross Profit	55%
Operating Expenses	35%
Net Income	12%

Balance Sheet

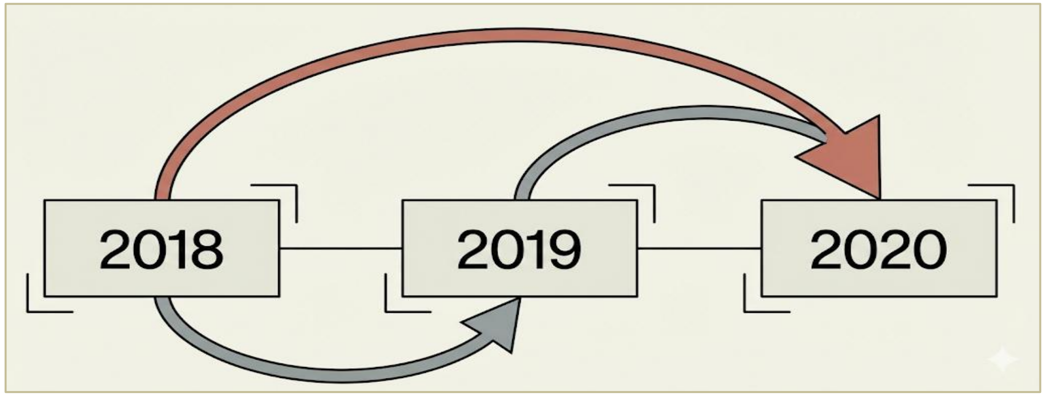
Base = Total Assets (100%)

Cash	15%
Receivables	20%
Inventory	25%
PP&E	40%
Total Assets	100%

Why Use Vertical Analysis?

Compare companies of **different sizes** • Analyze **cost structure** • Compare **across time** periods • Reveal **asset/liability composition**

Horizontal Analysis – Trend Analysis Over Time



- Identify trends over time
- Compares financial data across periods (year-to-year or quarter-to-quarter)

Two Measure of Change [ex. Revenue: $70,372 (yr 2) & $67,161 (yr 1)]

Dollar Change
Current Year – Prior Year
$70,372 – $67,161 = $3,211

Percent Change
(Current – Prior) / Prior
= 4.8%

Caution: Look at BOTH % and $ change — a tiny base can create misleading % changes!

Part 2

Return on Investment

ROE • ROA • ROFL • DuPont Model

Return on Equity (ROE)

$$\text{ROE} = \frac{\text{Net income}}{\text{Average stockholder's equity}}$$

Measures how effectively a company generates profit from shareholders' investment

Warren Buffett's Focus

- **High ROE & Low or reasonable debt**
- *High debt can artificially boost ROE*

Cautions

- ROE can be boosted by using more debt
- Excessive debt raises bankruptcy risk
- High ROE ≠ always good if driven by leverage

Use average balance (beginning + ending / 2) when comparing Balance Sheet amounts to Income Statement amounts

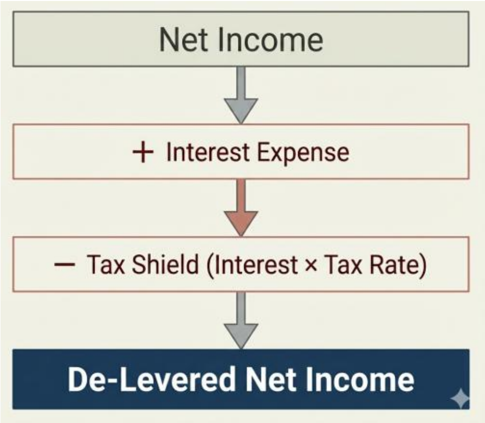
Return on Assets (ROA) – De-Levered ROA

$$\mathrm{ROE} = \frac{\text{Net income} + \text{Interest expense} \times (1 - \text{Tax rate})}{\text{Average Total Assets}}$$

Measures how efficiently a company uses its assets to generate profit — regardless of financing method

Why Add Back After-Tax Interest?

- Net income already includes interest expense.
- Interest is a financing cost, not an operational flaw. To measure pure asset performance, we remove the penalty of debt.
- This allows for comparison across companies regardless of their capital structure.

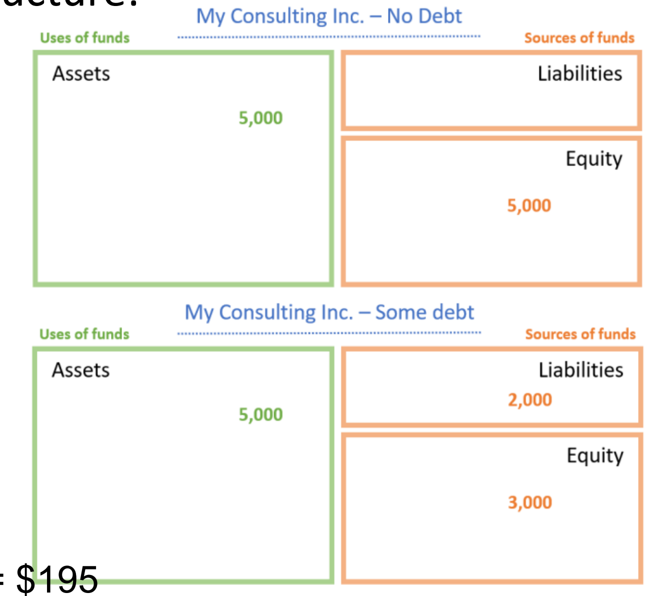


Standard ROA vs. De-Levered ROA

Term	**ROA (standard)**	**De-levered ROA**
Formula	$\frac{\text{Net Income}}{\text{Average total assets}}$	$\frac{\text{Net Income} + \text{after tax effect}}{\text{Average total assets}}$
Includes Debt Impact?	Yes	No
What It Shows	Profitability after interest cost (financing effects **included**)	Asset return from operations only (financing effects **excluded**)

Illustration: How de-levering NI removes the effects of capital structure:

	No debt	Some debt
Pretax, pre-interest income	$300	$300
Interest expense	0	(50)
Pretax income	300	250
Taxes (35%)	(105)	(87.5)
Net income	**195**	**162.5**
De-levered Net Income	**$195**	**$195**[a]



[a] after tax effect : NI+ Interest expense ×(1-Tax rate)= [162.5 + 50(1 - .35)] = $195

Return on Financial Leverage (ROFL)

ROFL = ROE – ROA

Measures how much of ROE is driven by financial leverage (use of debt) vs. operations

Key Concepts with PepsiCo 2020 Example:

ROE = 50.5% returns AFTER financing costs (includes debt effect)
ROA = 9.5% returns BEFORE financing costs (excludes debt effect, i.e., de-levered ROA)
ROFL = 41.0% the difference — shows leverage's contribution to shareholder returns

Interpretation:

Positive ROFL

Leverage increases ROE
Debt used effectively

Negative ROFL

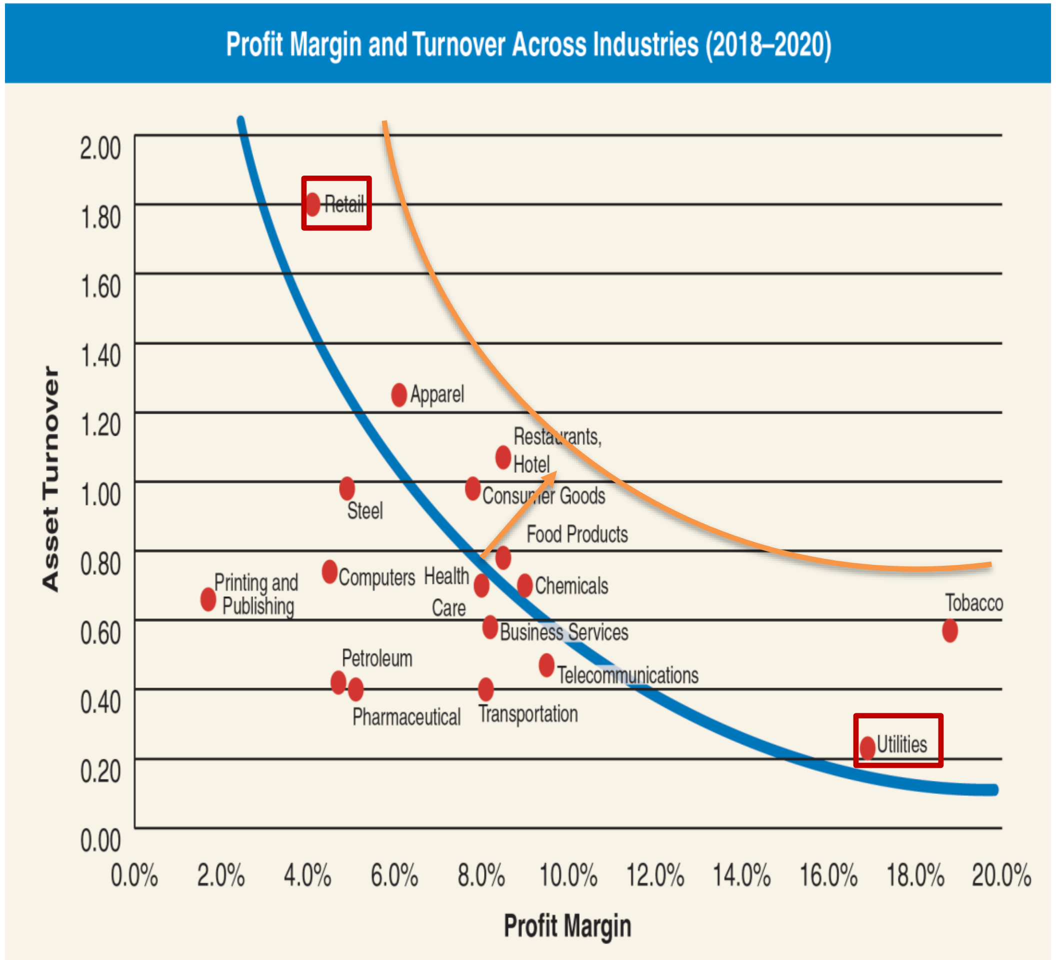
Leverage reduces ROE
Debt costs exceed benefits

Higher ROFL

= Higher Risk
Amplifies gains AND losses

Trade-Off Between Profit Margin & Asset Turnover

ROA = Profit Margin (Profitability) X Asset Turnover (Efficiency)



Why the Trade-Off Happens

- Firms achieve the **same ROA** through different PM–AT combinations
- **High PM strategies:** Premium pricing, lower volume, heavier asset investment (e.g., Utilities)
- **High AT strategies:** Low margins, high volume, efficient asset use (e.g., Retail)
- Movement **along** an iso-ROA curve reflects a change in strategy, not performance
- Shifts to **higher iso-ROA curves** indicate improved operating performance

Further Disaggregation of Profit Margin (PM)

Sales ($1.00)

COGS (Production Costs)	Operating Expenses (SG&A, R&D)	Operating Income (EBIT)

1. Gross Profit Margin (GPM)

(Sales - COGS) / Sales

- Reflects **product pricing**, **production efficiency**, and **cost control**.
- **PepsiCo 2020 : 54.8%**
  - 55¢ of every dollar is retained after making the product. High GPM funds R&D and Marketing.

2. Expense-to-Sales Ratio (ETS)

Operating Expenses / Sales

- Includes selling, general & administrative (SG&A), marketing, R&D, and advertising.
- **PepsiCo 2020: 40.5%**
  - 41¢ of every dollar goes to selling, admin, and overhead.

Sales	**$1.00**
— COGS	($0.45)
Gross Profit	**$0.55**
— SG&A — R&D — Marketing	($0.41)
Operating Income	**$0.04**
— Interest Expense	($0.01)
— Income Tax Exp	($0.01)
Net Income (Profit)	$0.02

ROA Disaggregation

$$ROA = \frac{\text{Earnings (NI or EWI)}}{\text{Average total assets}} = \frac{\text{Earnings}}{\text{Sales revenue}} \times \frac{\text{Sales revenue}}{\text{Average total assets}}$$

(Profit Margin) (Asset Turnover)

Profit Margin (PM)

Earnings / Net Sales

Measures: Profit per dollar of sales (Profitability)

Influenced by:

- Gross profit (price vs. COGS)
- Operating expenses
- Pricing power & cost control

Asset Turnover (AT)

Net Sales / Avg Total Assets

Measures: Sales per dollar of assets (Efficiency)

Driven by:

- Inventory management
- Receivables collection
- Asset intensity (industry)

ROA increases when a company earns strong profits (high PM) OR efficiently uses assets (high AT)

ROE Decomposition – The DuPont Model

ROE = Profit Margin × Asset Turnover x Financial Leverage

(Profitability) (Efficiency) (Use of Debt)

Profit Margin (PM)

Earnings / Net Sales

Measures: Profit per dollar of sales (Profitability)

Influenced by:

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$$\text{ROE} = \overbrace{\underbrace{\frac{\text{Net Income}}{\text{Sales}}}_{\substack{\text{Profit Margin}\\ \text{(Profitability)}}} \times \underbrace{\frac{\text{Sales}}{\text{Average Total Assets}}}_{\substack{\text{Asset Turnover}\\ \text{(Efficiency)}}}}^{\textbf{ROA}} \times \overbrace{\underbrace{\frac{\text{Average Total Assets}}{\text{Average Stockholders' Equity}}}_{\substack{\text{Financial Leverage}\\ \text{(Use of debt)}}}}^{\textbf{ROFL}}$$

Useful for identifying whether performance comes from:

- **Operational strength** (profitability, efficiency)
- Or **financial structure** (use of debt)

Limitation

- Uses net income (not EWI), so leverage affects both **Profit Margin** and **ROE**.
- Hard to isolate operating performance, but widely used.

ROE increases when a firm:

- Earns more profit per sales dollar (↑ PM) • Uses assets efficiently (↑ AT) • Employs more leverage (↑ FL)

How Debt Can Increase ROE

Two Engines of Return: Operations and Financing

ROE = ROA + ROFL

Operating Efficiency

Return on Assets generated by the core business.

Financing Strategy

Return on Financial Leverage generated by debt.

Why Debt Boosts ROE:

1. **Tax Shield:** Interest is tax-deductible (at 25% tax rate, $1 interest costs only $0.75)
2. **Lower Cost:** Debt is cheaper than equity (lenders have payment priority)
3. **Leverage Effect:** Generate more profit without raising new equity

Capital Structure and ROE

When Leverage Works FOR You

- **When business is performing well, leverage magnifies returns**
- **i.e., ROA (10%) > Interest Rate (4%)**

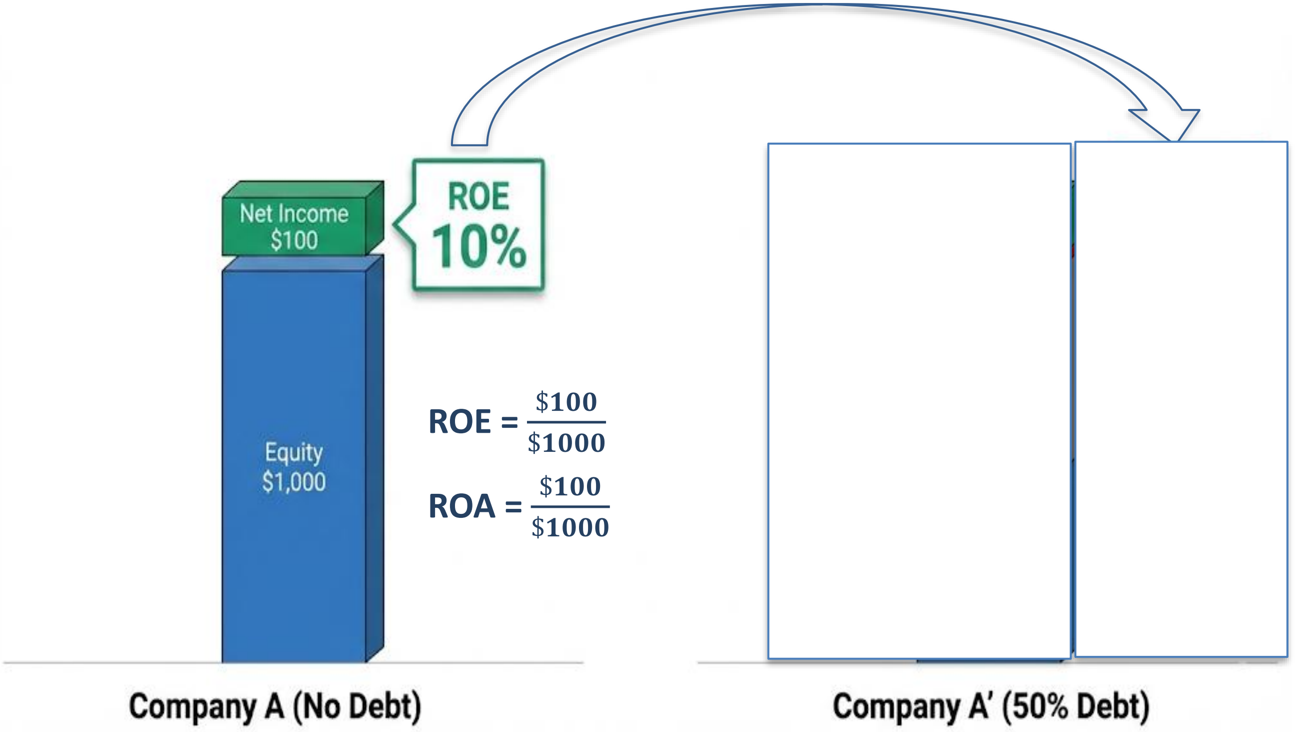
No Debt

ROE(10%) = ROA(10%) +ROFL(0%)

50% Debt

ROE(16%) = ROA(10%) +ROFL(6%)

** ROA is unlevered ROA.*



Capital Structure and ROE

When Leverage Works AGAINST You

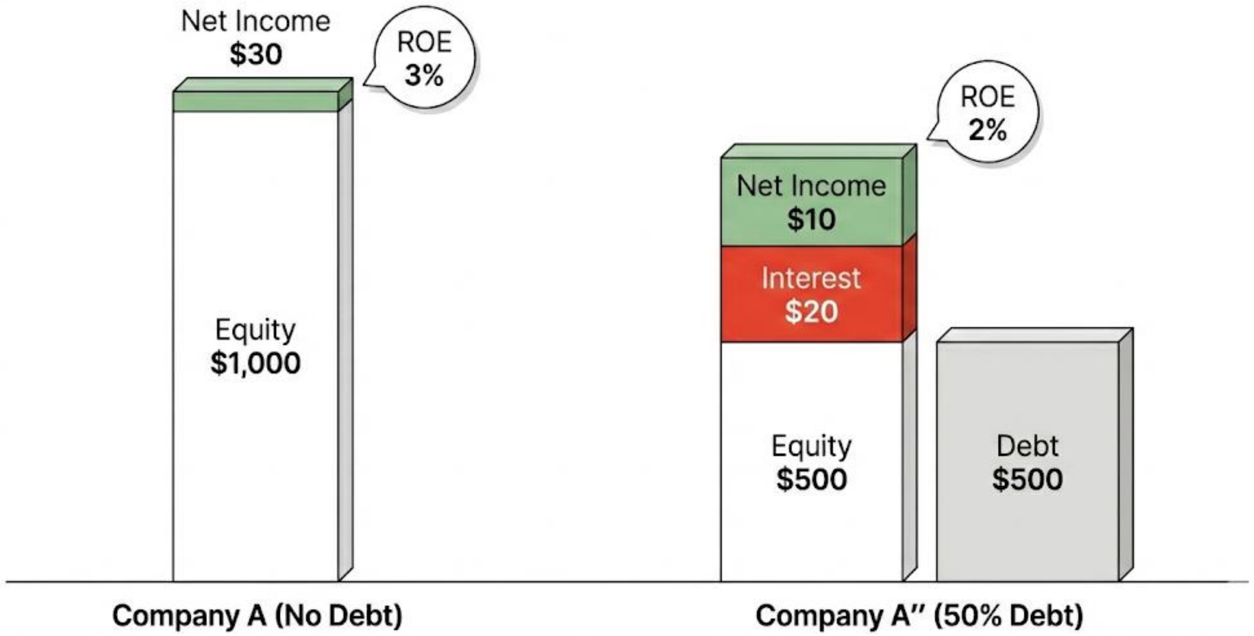
- **In underperforming markets, fixed interest payments erode value.**
- **i.e., ROA (3%) < Interest Rate (4%)**

No Debt

ROE(3%) = ROA(3%) +ROFL(0%)

50% Debt

ROE(2%) = ROA(3%) +ROFL(-1%)

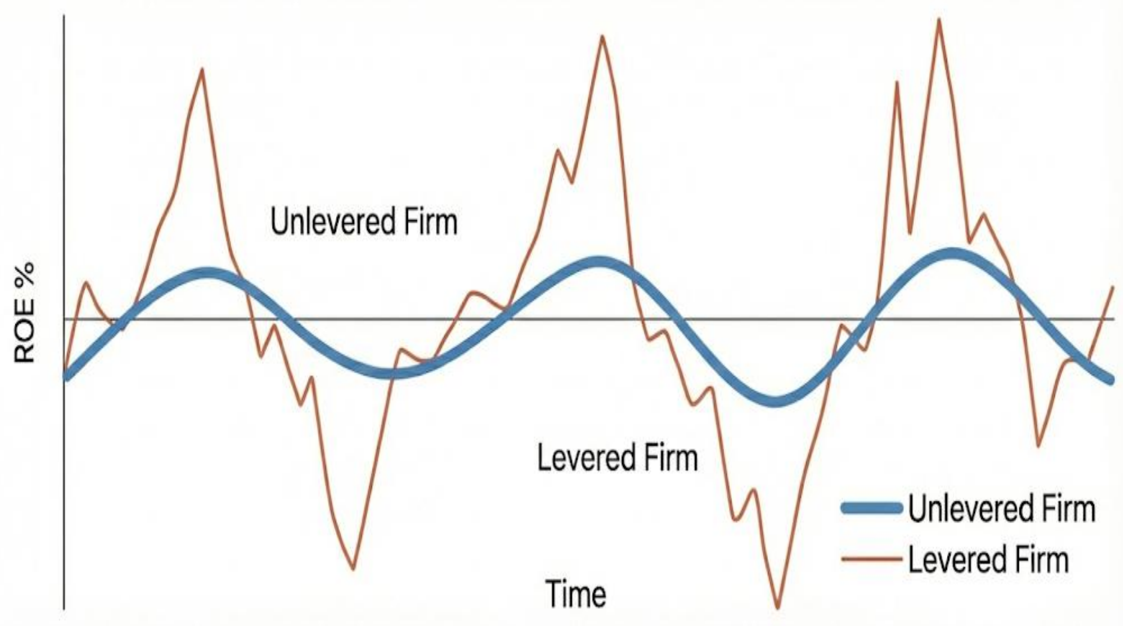


Downside Risk of Leverage

- Leverage does not just magnify returns.
- It magnifies volatility

The Cost of Leverage

- **Default Risk:** Higher chance of missing interest or principal payments (debt covenants violation)
- **Amplification:** Losses grow faster in downturns
- **Ultimate Cost:** Excessive debt can wipe out equity

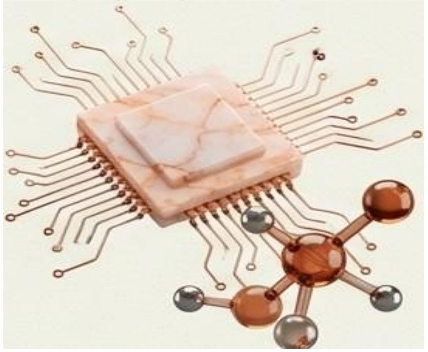


Matching Capital Structure to Strategy

High Debt Tolerance ← → **Low Debt Tolerance**

Utilities
Stable cash flows, hard assets

Telecom / Chemicals
Large fixed investments



Tech / Pharma
Intangible assets, volatile R&D

- Firms with predictable cash flows can safely carry more debt.
- Firms driven by innovation and intangibles must rely more on equity

Part 3

Liquidity & Solvency

Short-term and Long-term Financial Health

Liquidity Analysis

Can the company meet SHORT-TERM obligations?

Current Ratio

Current Assets / Current Liabilities

Above 1.0 = positive Working Capital

Quick Ratio

(Cash + AR) / Current Liabilities

Excludes inventory

OCF to CL

Operating Cash Flow / Current Liabilities

Operations can cover short-term debt?

Cash Burn Rate

Free Cash Flow / 365

How fast cash depletes

- Too low liquidity = default risk
- Too high liquidity = inefficient asset use

Solvency Analysis

Can the company meet LONG-TERM obligations?

Debt-to-Equity (D/E)

Total Liabilities / Stockholders' Equity

Measures: Reliance on debt vs equity

Higher D/E = More debt, lower solvency
Lower D/E = Conservative, more stable

Times Interest Earned (TIE)

(Operating Inc + Int Exp) / Int Exp

Measures: Can operating income cover interest?

Higher TIE = Comfortably covers interest
Lower TIE = May struggle to pay interest

Strong Solvency Position Means:

- Company can sustain debt levels and meet obligations over time
- Lower credit risk for lenders
- Greater borrowing capacity for future needs

Altman Z-Score (1968)

Altman Z-score predicts Bankruptcy Risk

$$\text{Z-Score} = 1.2(WC/TA) + 1.4(RE/TA) + 3.3(EBIT/TA) + 0.6(MVE/TL) + 1.0(Sales/TA)$$

Variables:

WC/TA: Working Capital / Total Assets → Liquidity
RE/TA: Retained Earnings / Total Assets → Long-term profitability
EBIT/TA: Operating profit / Total Assets → Efficiency
MVE/TL: Market Value Equity / Total Liabilities → Capital strength
Sales/TA: Sales / Total Assets → Asset turnover

Interpretation:

$Z > 3.0$	$Z = 1.8 - 3.0$	$Z < 1.8$
Healthy	Gray Zone	Distressed
Low bankruptcy risk	Moderate risk	High bankruptcy risk

Originally 95% accurate 1 year before bankruptcy. Still widely used by banks and credit analysts → ??

Working Capital Requirements

Traditional View – Working Capital as Liquidity

Working capital management deals with efficient control of cash and current assets to maintain liquidity

Formula

Net Current Assets = Current Assets – Current Liabilities

Interpretation

A positive figure indicates short-term solvency and operational stability

Key Insight

- Traditionally viewed as a liquidity measure — ensuring obligations can be met as they come due
- But this is only ONE way to view working capital...

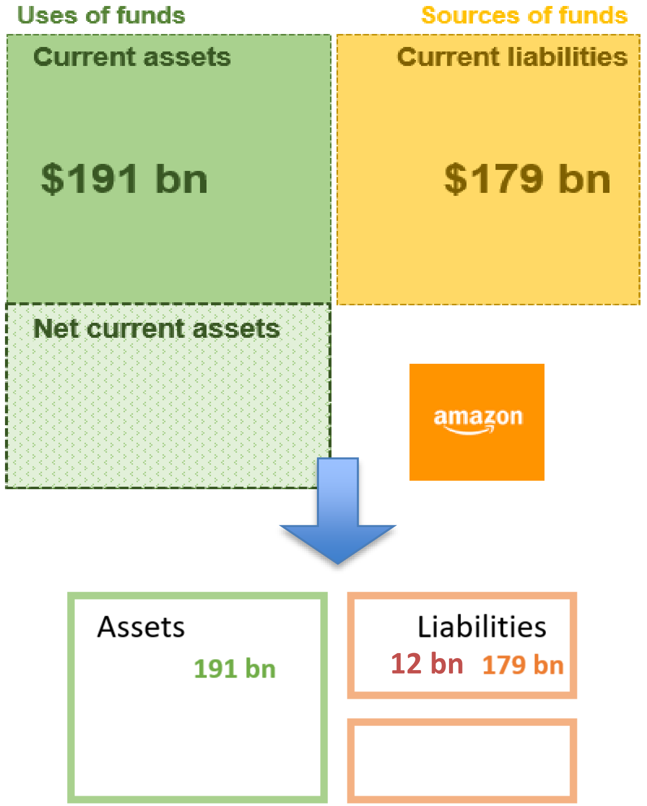
Source-of-Funds View –
The Working Capital Requirement

Concept Shift

Working capital can also be understood as the amount of long-term financing required to fund current operations.

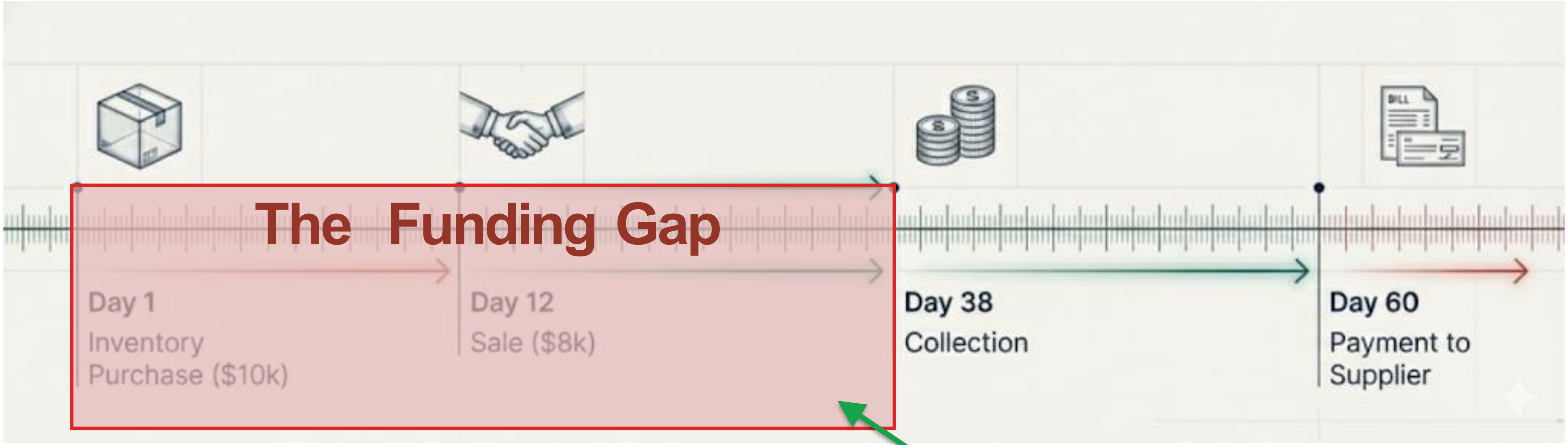
Key Points

- Net current assets create a working capital requirement [$12 bn (= $191 – $179)]
- It represents the portion of current assets NOT financed by current liabilities (or suppliers)
- The business must use long-term funds (equity or long-term debt) to cover this shortfall



When CA > CL, the gap must be financed by long-term capital — this is the true "working capital requirement."

The Cycle of Cash Lag



During this duration the business **requires long-term funding** to bridge the timing gap between inventory payments and cash collections.

Negative Working Capital: Risk vs. The 'Walmart' Advantage

What It Means

Occurs when current liabilities exceed current assets (CL > CA)

Implications – A Typical View (Risk)

- Short-term liabilities appear sufficient to fund current assets
- However, this position is often financially unhealthy and can signal liquidity risk
- The firm may struggle to meet short-term obligations

Remedies

- Increase current assets by selling non-current assets, or
- Raise additional long-term capital (equity or debt) to restore positive working capital

- Walmart Example: CA = $62 bn vs. CL = $77 bn → Net current liabilities of $15 bn.
- Walmart uses vendors' capital to finance operations. → Is it a bad thing?

Summary – Key Takeaways on Working Capital

Positive Working Capital: Healthy (Not Always Good)

- CA > CL → strong liquidity & solvency
- Too much WC = capital stuck in low-return assets

Working Capital = Funding Requirement

- Long-term capital needed to fund operations
- Not just liquidity — a capital allocation choice

Negative Working Capital: Risky (Not Always Bad)

- CL > CA → potential liquidity stress
- Can be efficient in some models
- *Example:* Walmart funds operations with supplier capital

Limitations of Ratio Analysis

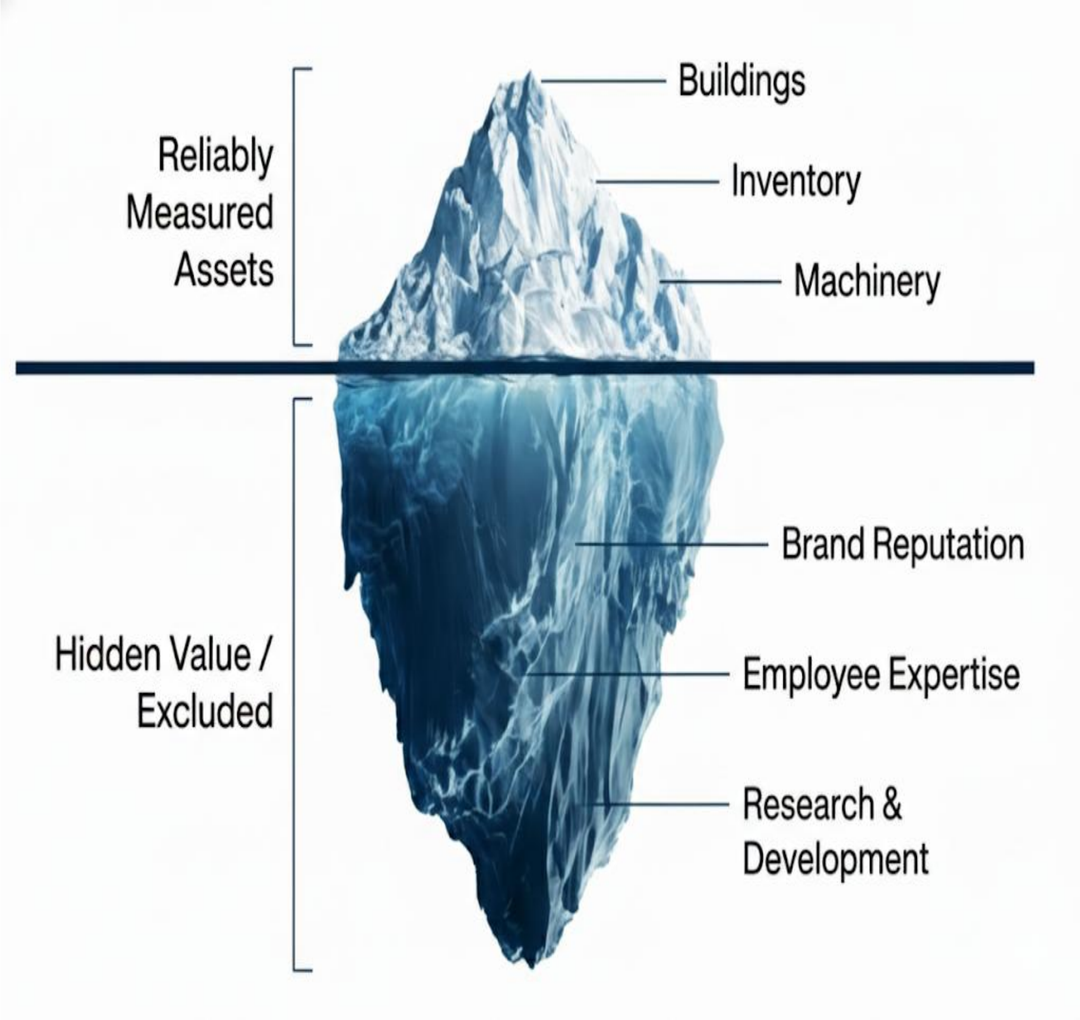
The Blind Spot 1: GAAP - Measurability

Measurability

- Financial statements include only items that can be **reliably measured**
- **Intangible but valuable assets** such as brand reputation, employee expertise, are often **excluded**, which can understate assets on the balance sheet
- Therefore**, ROA can be higher**

Non-Capitalized Costs

- Many expenditures that add long-term value (R&D, advertising) are expensed rather than capitalized
- This leads to understated assets on the balance sheet



The Blind Spot 2: Historical Cost vs. Market Reality

The Rule:
Assets are recorded at original acquisition cost

The Distortion:
Ratios based on these costs fail to reflect current market value, especially for long-held assets like real estate.

The Result:
Financial ratios can change simply because of accounting rules, not because performance.

An Example:
Land acquired at $1 million 20 years ago, but it is $10 million at today's market value.

Structural Distortions: The Comparability Problem

Company Changes

- Mergers, Acquisition and divestitures further complicate year-over-year comparison

Conglomerate Effects

- Large corporations operate across multiple industries.
- Consolidated statements obscure the specific risk and performance of individual segments

Timing Factors

- Seasonality and Life Cycle
- A start-up's ratios vs. mature "cash cow"

Example

- Amazon spans e-commerce, AWS, advertising, and entertainment.
- Consolidated ratios mix businesses with very different margins.
- Segment-level analysis is essential.

Ratios Are a Means to an End

Core Principle

- Ratios condense complex operations into single numbers.
- Ratios miss qualitative factors (management, innovation, strategy).
- Effective analysis looks beyond ratios to overall performance and direction.

What Ratios CAN tell You

- Trends in financial performance over time
- Relative positioning vs. industry peers
- Areas of potential concern (liquidity, leverage, profitability)
- Quick screening for investment analysis

What Ratios CANNOT Tell You

- Quality of management and corporate governance
- Strategic direction and competitive advantage
- Innovation pipeline and brand value
- Customer satisfaction and employee morale

Best Practice

- Use ratios as a starting point for deeper analysis — not as the final answer.

The Enlightened Analyst: Synthesis

Successful analysis requires looking beyond the ratios to understand the true funding architecture and strategic direction of the firm.

Data has Structural Limits

GAAP excludes intangibles and distorts values.

Working Capital is a Funding Requirement

Not just a liquidity check.

Ratios are a Means to an End

They cannot capture management, innovation, or strategy.

True insight lies in the gap between the number and the narrative.

Analyzing & Interpreting Core Operating Activities

Bifurcating the Business Model

The split: Every company runs two engines — one creates value, the other funds it.

Operating Activities

The Engine. Central transactions needed to produce and deliver goods or services.

Drivers: Sales, Production, R&D, Marketing.

Nonoperating Activities

The Support. Financing and investing activities that fund the business but are distinct from operations.

Drivers: Debt management, Investments.

Primary Value Driver

Secondary Driver

Why it matters: RNOA isolates the first engine so performance is not confused with financing.

The Metric of True Efficiency: RNOA

RNOA = Net Operating Profit After Taxes (NOPAT) ÷ Average Net Operating Assets (NOA)

Measures Return

Captures return generated strictly by operating activities.

Compares to Capital

Relates that return to the capital actually invested in those operations.

Removes Distortion

Strips out the distortions created by financing decisions.

In short: ROE mixes operating skill with financing choices; RNOA reports the operating engine on its own.

The Numerator: Purifying the Profit (NOPAT)

Goal: Start from Net Income, then strip out everything that is not operating.

Keep (Operating)

- Sales Revenue
- Cost of Goods Sold
- SG&A
- R&D
- Depreciation

Remove (Nonoperating)

- Interest Income
- Interest Expense
- Investment Gains / Losses

Result — NOPAT: the profit the business would earn with no debt and no investment income.

Calculating NOPAT

NOPAT = Net Income − (Nonoperating Revenues − Nonoperating Expenses) × (1 − Tax Rate)

Key Insight

Why this formula? We are effectively adding back the after-tax cost of debt.
We want to see the profit the company would report if it had no debt and no investment income.

Nonoperating Revenues

Interest and investment income — remove it, it is not from operations.

Nonoperating Expenses

Interest cost of debt — add it back, it is a financing choice.

Case Study: PepsiCo (2020)

Objective: Determine the efficiency of the core business, separate from its capital structure.

Net Income	Nonoperating Revenues	Nonoperating Expenses	Tax Rate
$7,175M	$117M	$1,128M	25%

Note: All figures in millions of USD. The 25% rate is the effective rate used in the NOPAT adjustment on the next slide.

Why adjust: Net Income alone reflects PepsiCo’s financing mix. NOPAT and NOA remove it.

PepsiCo: The NOPAT Adjustment

Net Income	Less: After-Tax Nonoperating Net	NOPAT
$7,175.0	−$758.3	$7,933.3

($117 − $1,128) × (1 − 0.25) = −$758.3 → $7,175.0 − (−$758.3) = $7,933.3

Observation

NOPAT is higher than Net Income. By removing the “drag” of interest payments, we see the true earning power of the operations — the profit available to all capital providers, debt and equity alike.

The Denominator: Capital at Work (NOA)

Question: Split the balance sheet — which items actually run the business?

Operating Assets

Cash • Receivables • Inventory
PP&E • Intangibles

Operating Liabilities

Accounts Payable • Accrued Expenses
Employee Benefits

Nonoperating Assets

Investments in affiliates and marketable securities.

Nonoperating Liabilities & Equity

Debt and interest-bearing liabilities • Equity.

NOA = Operating Assets − Operating Liabilities

PepsiCo: Isolating Operating Capital

Total Operating Assets	Minus Operating Liabilities	Net Operating Assets (NOA)
$91,059	−$35,216	$55,843

$91,059 − $35,216 = $55,843

What This Number Means

This $55.8 billion is the equity and debt capital actively funded into the machinery, inventory and brand of PepsiCo — the capital the operating engine is actually working with.

The Verdict: PepsiCo's Operating Efficiency

RNOA = NOPAT ÷ Average NOA = $7,933.3 ÷ $51,082.5 = 15.5%

NOPAT

$7,933.3 million

Average NOA

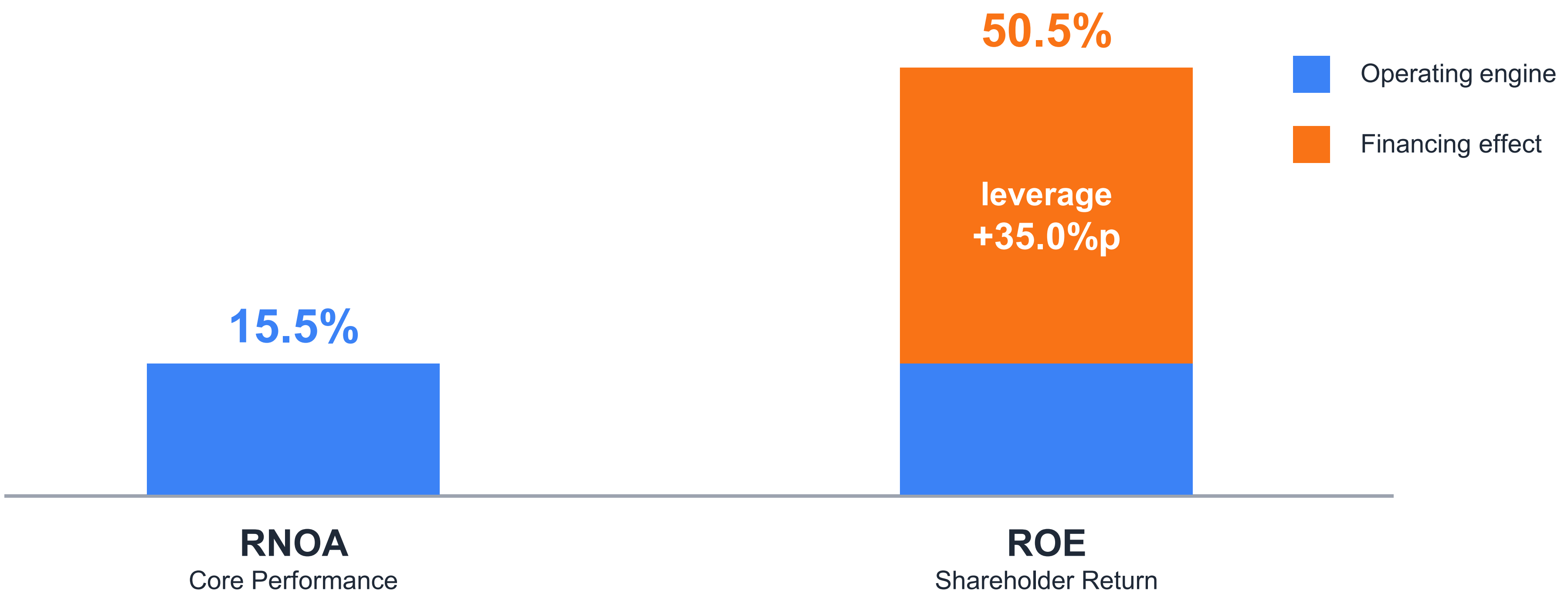
($55,843 + $46,322) ÷ 2
= $51,082.5 million

RNOA

15.5%

Strong operating efficiency: For every dollar invested in core operations, PepsiCo generates 15.5 cents of after-tax profit.

Signal vs. Noise: RNOA vs. ROE



Reading the Gap

Most of the shareholder return is engineered through leverage (nonoperating), not by selling more product (operating). Use RNOA to judge the business; use ROE to judge the capital structure.

The Analyst's Checklist

1

Distinguish

Separate operating (value creation) from nonoperating (financing).

2

Adjust Income

Convert Net Income to NOPAT by removing interest and investment noise.

3

Adjust Capital

Convert Total Assets to NOA by isolating working capital and PP&E.

4

Interpret

Use RNOA to judge the business manager; use ROE to judge the capital structure.

The two questions: RNOA answers “how good is the business?” ROE answers “how good is the business plus its financing?”

Clarity in Capital

Operating activities are central to value creation.

The Payoff

By isolating them, we move from simply reading the statements to understanding the business engine.

Distinguish

Operating vs. nonoperating

Adjust

NOPAT and NOA

Interpret

RNOA vs. ROE